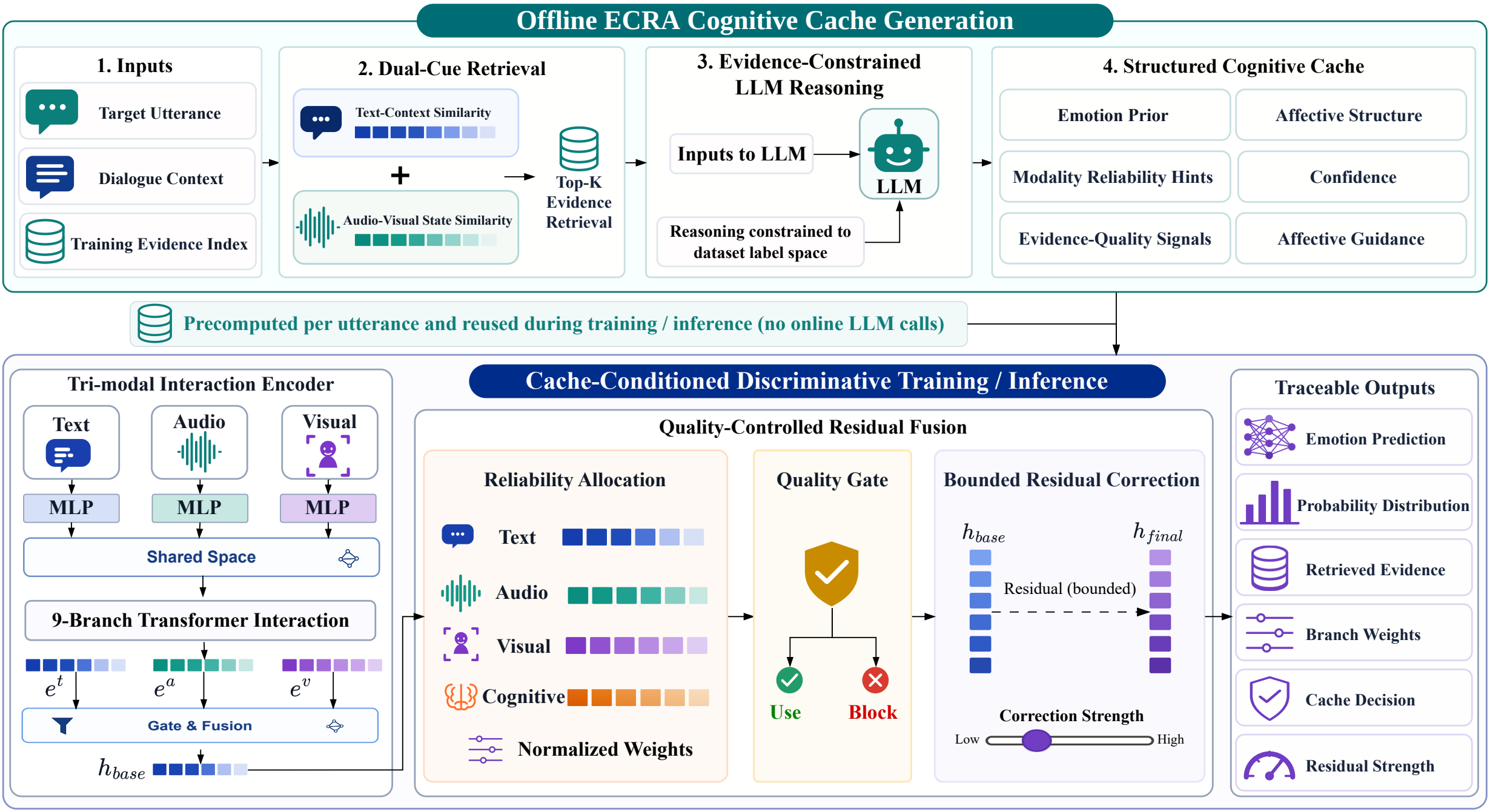

Offline ECRA Cognitive Cache Generation
1. Inputs
Target Utterance
Dialogue Context
Training Evidence Index
2. Dual-Cue Retrieval
Text-Context Similarity
Audio-Visual State Similarity
Top-K Evidence Retrieval
3. Evidence-Constrained LLM Reasoning
Inputs to LLM
LLM
Reasoning constrained to dataset label space
4. Structured Cognitive Cache
Emotion Prior
Affective Structure
Modality Reliability Hints
Confidence
Evidence-Quality Signals
Affective Guidance
Precomputed per utterance and reused during training / inference (no online LLM calls)
Tri-modal Interaction Encoder
Text
Audio
Visual
MLP
MLP
MLP
Shared Space
9-Branch Transformer Interaction
e^t
e^a
e^v
Gate & Fusion
h_{base}
Cache-Conditioned Discriminative Training / Inference
Quality-Controlled Residual Fusion
Reliability Allocation
Text
Audio
Visual
Cognitive
Normalized Weights
Quality Gate
Use
Block
Bounded Residual Correction
h_{base}
h_{final}
Residual (bounded)
Correction Strength
Low
High
Traceable Outputs
Emotion Prediction
Probability Distribution
Retrieved Evidence
Branch Weights
Cache Decision
Residual Strength